

Adult guide — E03 The Fold Makes a Pattern

Version: review 1.0.0 Story: The Turning-Light Trail Author: Maria Smith Status: working review; not an approved final publication

What this book is for

E03 is a read-aloud story and activity book for Early Years / Foundation. It continues directly from E02. The rebuilt Moon Lantern begins to show two lights, and the child helps Mia, Sol, Tavi and one new guest, Vee, repair the route to the Sunrise Arch.

The child should increasingly be able to:

- keep two visible roles distinct without calling one positive and one negative;
- describe what one declared move does;
- notice that the next lawful move brings the earlier role back;
- copy and continue a short declared recurrence;
- retain and identify the first move that breaks the rule; and
- use the same two-role rule after the names and pictures change.

This is not a conventional pattern book with Fold vocabulary added. The topic and sequence come from the admitted Foundation dependency spine.

Opening recap and invitation

Book Three and Level Three start by joining the story back together before the new problem appears. The opening must be understandable whether the child has played the earlier levels recently, read only the books, or chosen Level Three first from the level screen.

The book opening and the game's short prelude should:

1. welcome the child to Mia, Sol and Tavi in the Star Door world;
2. recall that the team took the Moon Lantern apart, carried all four parts through the small door, counted each part once and rebuilt the same whole;
3. show the complete lantern safely on the balcony before it begins changing;
4. let the child see its light turn from blue moon to gold sun and back, without naming the rule too early;
5. show that the changing light wakes a trail towards the Sunrise Arch, but the trail stops before the team can reach it;
6. introduce Vee as the one new friend and state the mission in simple words: "Help Vee restore the turning-light trail before sunrise"; and
7. tease that the child will turn, compare, continue and repair the trail, without showing the correct next role or route.

This prelude is a visual story scene rather than an instruction list. A returning child may replay or skip it, but skipping must not complete the first activity or reveal any answer.

Scientific authority and exact boundary

SFT - FOUNDATION - FOLD - 001

- Receipt:
sha256:9e99f91d15e2b68a5c24f825093686863c168595319c2083e6a5b52b0314e003
- Status: model admitted, depth-independent, independently replicated
- Exact role in E03: the minimal parameter-free Fold has two complete, disjoint, equal fibres with distinct held labels and an explicit complete return relation to the One.

The blue moon and gold sun faces translate the experience of retaining two distinct roles. They are not the proof labels and the cardboard tile is not the Fold theorem.

SFT - FOUNDATION - FOLD - DYNAMICS - 001

- Receipt:
sha256:ffaa7c3740dd747aecfc858391cec140dd6c1b3c8ab219ca798fccfc1785b675
- Status: model admitted, depth-independent, independently replicated
- Exact role in E03: exact Fold dynamics retains two fibre roles, gives a declared transition, identifies the return relation and contains the first nonstatic recurrent orbit.

The book translates only the two-role turn-and-return action. It does not teach the claim's rational part arithmetic, exact doubling-and-cast equation, fibre witnesses or uniform even divisions at child level.

What the activities do not prove

A hand turn, paper card, coloured light, over/under path or star/leaf row is a bounded demonstration. None independently derives the admitted claims. The book does not assert that every alternating pattern is an SFT Fold or that a physical flip supplies scientific authority.

The following remain excluded:

- binary digits, bit axioms or Boolean authority;
- signed or positive/negative numerical labels;
- semantic numerical zero;
- floating-point equality;
- completed infinity or an endless completed pattern;
- conventional probability or physical phase as a premise; and
- any answer selected because it is familiar from another curriculum.

Story cause-chain

Keep the mission audible: restore the turning-light trail before sunrise.

- Pages 3–5: the alternating Moon Lantern light starts the trail, the trail stops, and Vee asks for help.
- Pages 6–8: the child experiences two distinct sides before their role names are used independently.

- Pages 9–14: one turn changes the visible side; the next turn returns the first side.
- Pages 15–17: the child copies the actions before meeting the word pattern.
- Pages 18–21: Sol's failed attempt stays visible; the child finds and repairs the first broken move.
- Pages 22–27: over/under transfers the same declared two-role rule and the child chooses the complete lawful route.
- Pages 28–31: delayed recall tests the first transition; star/leaf tests transfer after the role names change.
- Page 32: the preserved failed attempt creates the question for E04.

If a child loses the story, restate the mission and point to the lit trail and dark trail. Do not jump straight to the next answer.

Challenge and reveal answers

- Page 7 — Which side is showing? Inspect Vee's tile and both answer pictures. The reveal shows the blue moon side.
- Page 10 — What does one turn show? Turn a flat hand or safe card once. The reveal shows the gold sun.
- Page 12 — What happens after one more turn? Turn the hand or card again. The reveal shows the blue moon returning.
- Page 16 — Which light comes next? Point through all four shown lights. The reveal shows a blue moon.
- Page 19 — Where does the rule first break? Check each move from left to right. The fourth blue moon is the first break; a gold sun is required.
- Page 23 — Which bridge role comes next? Say “over, under, over” slowly. The reveal shows “under”.
- Page 26 — Which route is complete and keeps the rule? Inspect every adjacent move and where each route ends. The reveal shows Route C.
- Page 28 — What came out of the first gate? Recall the first gate before looking back. The reveal shows the gold sun.
- Page 30 — Which new role comes next? Say “star, leaf, star” without using colour cues. The reveal shows a leaf.

Wrong answers are useful records. Ask “What is showing now?” and “What does one move do?” Preserve the chosen picture until the child has compared it. Do not say “No, that is wrong” or praise a child as clever for the answer.

What the narrator explains after each game

After every mini-game, the narrator speaks to the child directly. Each short explanation names the action just completed, the visible result, the idea the child has discovered, and why that idea helps repair the trail. It comes after an attempt and does not turn the next activity into a told answer.

1. Catch the lantern lights: the child caught the moving moon and sun lights and found where the trail stopped. Looking at which lights are lit and which are dark gives the

team a clear starting record.

2. Inspect both sides: one tile has a blue moon side and a gold sun side. The sides are different roles of the same tile; neither side has disappeared while the other is showing.
3. Turn at the first gate: one declared turn changes the visible role from blue moon to gold sun. A turn means this one agreed move, so the team can check its result.
4. Find the returning path: one more lawful turn brings the blue moon back. Return means the earlier role shows again after the next agreed move.
5. Continue the light trail: repeating the same turn gives moon, sun, moon, sun. A pattern is the stated rule that tells the team which lawful move comes next, not merely a row that looks familiar.
6. Keep and repair Sol's try: checking from the first move finds the earliest place the turn rule breaks. Keeping Sol's first route lets the child compare the change and repair only the broken move.
7. Guide Vee over and under: five arches alternated over, under, over, under, over, or the reverse from an UNDER start. The same two-role rule worked with new names.
8. Choose the complete route: the accepted route follows the rule at every move and reaches the Sunrise Arch. Checking the whole route matters because a good first step alone does not complete the journey.
9. Continue new roles and remember the first gate: star and leaf can follow the same two-role rule, and recalling the first gate checks whether the child understood the move rather than copied the last picture.

The final fourth-wall explanation should tell the child: the team discovered two different roles and one clear move that changes which role shows. Repeating the move brought the first role back and made a rule the child could continue, check and repair. The same rule still worked when moon and sun became over and under, then star and leaf. This matters because a clear rule helps us predict what comes next, find the first change that does not fit, and try a careful repair. The spoken line and its visible caption must match exactly.

Plain-language meanings

- Side: one named face of the same tile.
- Turn: one declared move that changes which named side shows.
- Return: the next lawful move brings the earlier side back.
- Pattern: a declared rule that tells us what move comes next.
- Broken rule: one visible move does not follow the declared rule.
- Role: the place a named picture has in the declared rule. The pictures may change while the two-role rule stays the same.

Do not use held label, fibre, phase, antipode, orbit or two-cycle as unexplained child shorthand.

Safe physical activities

Use only a large card, cloth square or adult hand. Do not use small loose parts with children who may mouth them. The child may:

- draw a moon on one side of a card and a sun on the other;
- turn the card on a table rather than throwing it;
- step over and under an imaginary line with clear space and adult support; or
- point to large printed roles instead of moving physically.

Every activity has a pointing or spoken alternative. Movement is optional.

Accessibility

- Blue moon and gold sun are distinguished by symbol and name as well as colour.
- Over and under use words and route position, not colour.
- Challenge choices remain visible until the labelled reveal.
- All image descriptions name the current role, the move and the location of a broken move.
- The accessible HTML edition must preserve page order and expose every picture description beside its page text.
- The game must work by touch, keyboard and assistive technology and must honour reduced-motion settings.
- Local narration must match the visible caption exactly; sound effects are not required to understand an answer.

Companion Level Three contract

Level Three must retain Mia, Sol and Tavi as the permanent team and introduce only Vee. It opens with the recap and story invitation above, then includes eight or more story stages, each with a distinct, replayable mini-game:

1. catch moving lantern lights and reveal the stopped trail;
2. inspect both sides of one tile and record both pictures;
3. pull the first gate handle to its gold mark, with a tap alternative;
4. compare complete paths and choose the path that returns the moon;
5. choose between two visibly moving lights and place the next three lawful lights;
6. preserve, identify and repair the first broken move;
7. guide Vee through five alternating over-and-under arches;
8. inspect three complete route maps and choose the lawful route; and
9. continue star-and-leaf roles from one fixed clue, then recall the first gate's result.

The game is not a menu of questions. Characters move in the scene, each action changes the visible trail, progress restores exactly after tab or page changes, and every completed mini-game offers replay.

Narration includes the opening recap and every post-game explanation. It is pre-rendered with Maria Smith's local Kokoro voices, works offline, and matches the visible caption exactly. A finished line must not replay merely because the child changes tabs or returns to the app.

Level Three has its own gentle travelling rhythm, shaped by alternating musical pulses that support the turning-light journey without giving an answer. It is distinct from Level One's magical mystery theme and Level Two's workshop theme. The music stays below voices and softens during narration. A plainly labelled music control turns background music on or off separately from narration and action sounds and remembers the setting on the device. No musical cue may be required to solve a game. Leaving the app pauses all audio; returning restores only the current scene without overlapping or restarting completed speech.

Connecting the book and Level Three

The book and game tell the same causal story and teach the same rule. The book slows the child down to compare challenge and reveal; the game lets the child perform, repair and replay the moves. Neither format is homework for the other.

- Begin with the mission: “The turning-light trail stopped. We are helping Vee repair it before sunrise.”
- Before a move, ask “What is showing now?” and “What does one turn do?” Do not say which picture or route comes next.
- Use a book challenge before or after its matching mini-game. Once the child has tried and heard the narrator's explanation, invite them to show the same rule in the other format.
- If the child is stuck, trace the completed moves from the beginning or let them operate one move again. Do not point to the next answer.
- Keep the first broken route visible and compare it with the repair. The aim is to understand the rule, not to hide evidence of a useful attempt.
- Optional codes add animation, character moments or replay variety only. They do not reveal the next role, route or remembered answer.

Optional book codes

Codes never reveal an answer or skip a learning action.

- TRAILLIGHT — page 5: the Sunrise Arch plays a short two-light greeting.
- TWOSIDES — page 8: Vee's goggles reflect both role symbols.
- TURNBACK — page 14: a slow-motion turn-and-return replay.
- NEXTLIGHT — page 17: Sol performs a two-step light dance after the game is complete.
- KEEPTHE TRY — page 21: the preserved first-try card receives a gold frame.
- OVERUNDER — page 24: the bridge plays two different route notes.
- RIGHTROUTE — page 27: a cosmetic light trail appears after the route is solved.
- NEWROLES — page 31: a small preview of E04's checking desk.

External references

The UK Early Years Foundation Stage framework may calibrate activity length, communication demands and accessibility. It is not authority for SFT claims or their dependency order. Publisher styles, conventional curricula and other games may inform usability comparisons only.

Review and publication status

Review version 1.0.0 remains under edu/. It may be rendered and tested as a review work. It must not be copied into publications/education, presented as approved or remotely hosted as a child service until Maria Smith explicitly approves the exact final book and companion level.